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STAT2920: Introduction to Probability

Tutorial for Chapter 6: Special Probability Densities

Exercise 1.
Show that if a random variable has a uniform density with the parameters α and β, the r^{th} moment about the mean equals

(a) 0 when r is odd;

(b) $\frac{1}{r+1}\left(\frac{\beta-\alpha}{2}\right)^r$ when r is even.

Solution 1.
When $X \sim \text{Unif}(\alpha, \beta)$, the r^{th} moment about the mean is given by

$$\begin{aligned}
\mu_r = \frac{1}{\beta-\alpha}\int_\alpha^\beta \left(x - \frac{\alpha+\beta}{2}\right)^r \mathrm{d}x &= \frac{1}{(\beta-\alpha)2^r}\int_\alpha^\beta \big(2x-(\alpha+\beta)\big)^r \mathrm{d}x \\
&= \frac{1}{(\beta-\alpha)2^r}\left(\frac{\big(2x-(\alpha+\beta)\big)^{r+1}}{2(r+1)}\right)\Bigg|_\alpha^\beta \\
&= \frac{1}{(\beta-\alpha)2^r}\cdot\frac{(\beta-\alpha)^{r+1}-(-1)^{r+1}(\beta-\alpha)^{r+1}}{2(r+1)}
\end{aligned}$$

(a) $= 0$ when r is odd.

(b) $= \frac{1}{(\beta-\alpha)2^{r+3}(r+1)}\cdot 2(\beta-\alpha)^{r+1} = \frac{1}{r+1}\left(\frac{\beta-\alpha}{2}\right)$ when r is even.

Exercise 2.
Suppose that Y is the working life (in years) of a brand new industrial machine. The following is the probability density function of Y.

$$f(y) = \frac{1}{2}\left(\frac{1}{5}\right)^3 y^2 e^{-\frac{y}{5}}, \quad y > 0$$

A manufacturing plant has just purchased such a new machine. Determine the probability that the machine will be in operation for the next 20 years.

Solution 2.

First, note that $Y \sim \text{Gamma}(\alpha = 3, \beta = 5)$, the requested probability is given by

$$\begin{aligned}
\mathbb{P}(Y \geq 20) &= 1 - \mathbb{P}(Y < 20) \\
&= 1 - \int_0^{20} f(y)\,\mathrm{d}y \\
&= 1 - \int_0^{20} \frac{1}{2}\left(\frac{1}{5}\right)^3 y^2 e^{-\frac{y}{5}}\,\mathrm{d}y \\
&\approx 0.2381 \quad \text{apply integration by parts}
\end{aligned}$$

Exercise 3.

Suppose that a random variable X has a probability density function given by

$$f(x) = \begin{cases} cx^3 e^{-\frac{x}{2}}, & x > 0 \\ 0, & \text{elsewhere.} \end{cases}$$

Find the constant c

Solution 3.

We recognize the gamma distribution with parameters $\alpha = 4$ and $\beta = 2$, then

$$c = \frac{1}{\Gamma(\alpha)\beta^\alpha} = \frac{1}{\Gamma(4)2^4} = \frac{1}{96}.$$

Exercise 4.

Suppose that the length of a phone call in minutes is an exponential random variable with parameter $\lambda = 1/10$. If someone arrives immediately ahead of you at a public telephone booth, find the probability that you will have to wait

(a) more than 10 minutes;

(b) between 10 and 20 minutes

Solution 4.

Let X denote the length of the call made by the person in the booth. Then the desired probabilities are

(a)

$$\begin{aligned}
\mathbb{P}(X > 10) &= 1 - \mathbb{P}(X \leq 10) \\
1 - F(10) &= 1 - (1 - e^{-1}) = e^{-1} \approx 0.368
\end{aligned}$$

(b)

$$\begin{aligned}
\mathbb{P}(10 < X < 20) &= F(20) - F(10) \\
e^{-1} - e^{-2} &\approx 0.233
\end{aligned}$$

Exercise 5.

The number of days that elapse between the beginning of a calendar year and the moment a high-risk driver is involved in an accident is exponentially distributed. An insurance company expects that 30% of high-risk drivers will be involved in an accident during the first 50 days of a calendar year.

Calculate the portion of high-risk drivers are expected to be involved in an accident during the first 80 days of a calendar year.

Solution 5.

Let X denote the number of days that elapse before a high-risk driver is involved in an accident. Then X is exponentially distributed with unknown parameter λ. We are given that

$$0.3 = \mathbb{P}\left[X \leq 50\right] = \int_0^{50} \lambda e^{-\lambda x}\,\mathrm{d}x = -e^{-\lambda x}\Big|_0^{50} = 1 - e^{-50\lambda}$$

Therefore, $e^{-50\lambda} = 0.7$ and $\lambda = -\frac{1}{50}\ln(.7)$. Then,

$$\begin{aligned}\mathbb{P}\left[X \leq 80\right] &= \int_0^{80} \lambda e^{-\lambda x}\,\mathrm{d}x = -e^{-\lambda x}\Big|_0^{80} = 1 - e^{-80\lambda} \\ &= 1 - e^{(80/50)\ln(0.7)} = 0.435.\end{aligned}$$

Exercise 6.

Let X be a gamma random variable $\mathsf{Gamma}(\alpha, \beta)$ with mean $E(X) = 20$ and $Var(X) = 100$. Find α and β, the two parameters.

Solution 6.

One has

$$\alpha\beta = 20, \qquad \alpha\beta^2 = 100.$$

Then

$$\beta = \frac{100}{20} = 5, \qquad \alpha = \frac{20}{5} = 4.$$

$\square$

Exercise 7.

If the annual proportion of erroneous income tax returns filed with the IRS can be looked upon as a random variable having a beta distribution with $\alpha = 2$ and $\beta = 9$, what is the probability that in any given year there will be fewer than 10 percent erroneous returns?

Solution 7.

Let X be the annual proportion of erroneous income tax, given that $X \sim \text{Beta}(\alpha = 2, \beta = 9)$, then

$$\begin{aligned}
\mathbb{P}[X \leq 0.1] &= \frac{1}{B(2,9)} \int_0^{0.1} x^{2-1}(1-x)^{9-1}\, \mathrm{d}x \\
&\overset{y=1-x}{=} 90 \int_{0.9}^{1} y^8(1-y)\, \mathrm{d}y \\
&= 90 \left(\frac{y^9}{9} - \frac{y^{10}}{10} \right)\Bigg|_{0.9}^{1} \\
&= 90 \left(\frac{1}{9} - \frac{1}{10} - \frac{(0.9)^9}{9} + \frac{(0.9)^{10}}{10} \right) = 0.264
\end{aligned}$$

Exercise 8.

Determine the constant C such that the function

$$f(x) = Cx^3(1-x)^6, \qquad 0 \leq x \leq 1$$

is a beta distribution. Also, find its mean and variance.

Solution 8.

One has

$$\begin{aligned}
&\int_0^1 Cx^3(1-x)^6 dx = 1 \\
\Longrightarrow\ &\frac{1}{C} = \int_0^1 x^{4-1}(1-x)^{7-1} dx \\
\Longrightarrow\ &C = \frac{1}{B(4,7)} = \frac{\Gamma(4+7)}{\Gamma(4)\Gamma(7)} = \frac{10!}{3!6!} = 840.
\end{aligned}$$

Then

$$E[X] = \frac{4}{4+7} = \frac{4}{11}, \qquad Var(X) = \frac{4 \cdot 7}{(4+7)^2(4+7+1)} = \frac{28}{(121)(12)} = \frac{7}{363}.$$

$\square$

Exercise 9.

If Z is a random variable having the standard normal distribution, use the z-table at the end to find

(a) $\mathbb{P}(Z < 1.33)$;

(b) $\mathbb{P}(Z \geq -0.79)$;

(c) $\mathbb{P}(0.55 < Z < 1.22)$;

(d) $\mathbb{P}(-1.90 \le Z \le 0.44)$.

Solution 9.

Using the tables in the next pages, we get

(a) $\mathbb{P}(Z < 1.33) = 0.9082$

(b) $\mathbb{P}(Z \geq -0.79) = 1 - \mathbb{P}(Z < -0.79) = 1 - 0.2148 = 0.7852$

(c) $\mathbb{P}(0.55 < Z < 1.22) = \mathbb{P}(Z < 1.22) - \mathbb{P}(Z < 0.55) = 0.8888 - 0.7088 = 0.1800$

(d) Similarly, we get $\mathbb{P}(-1.90 \le Z \le 0.44) = \mathbb{P}(Z < 0.44) - \mathbb{P}(Z < -1.90) = 0.6700 - 0.0287 = 0.6413$

Exercise 10.

The life (in hours) of an electronic tube manufactured by a certain process is normally distributed with mean 160 and variance σ^2. What is the maximum allowable value for σ if the life of a tube is to have a probability 0.8 of being between 120 and 200 hours? (Leave the answer in terms of the inverse the standard normal CDF Φ.)

Solution 10.

Let X be the life (in hours) of a randomly chosen tube. Then by assumption $X \sim \mathcal{N}(160, \sigma^2)$. Then

$$
\begin{aligned}
& P(120 \le X \le 200) = 0.8 \\
\implies & P\left(\frac{120-160}{\sigma} \le \frac{X-160}{\sigma} \le \frac{200-160}{\sigma}\right) = 0.8 \\
\implies & P\left(-\frac{40}{\sigma} \le Z \le \frac{40}{\sigma}\right) = 0.8 \\
\implies & \Phi\left(\frac{40}{\sigma}\right) - \Phi\left(-\frac{40}{\sigma}\right) = 2\Phi\left(\frac{40}{\sigma}\right) - 1 = 0.8 \\
\implies & \Phi\left(\frac{40}{\sigma}\right) = 0.9 \\
\implies & \frac{40}{\sigma} = \Phi^{-1}(0.9) \implies \sigma = \frac{40}{\Phi^{-1}(0.9)} (\text{hrs}).
\end{aligned}
$$

$\square$

	Standard Normal Distribution Table									
	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

	Standard Normal Distribution Table									
	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-1	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-2	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-3	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010